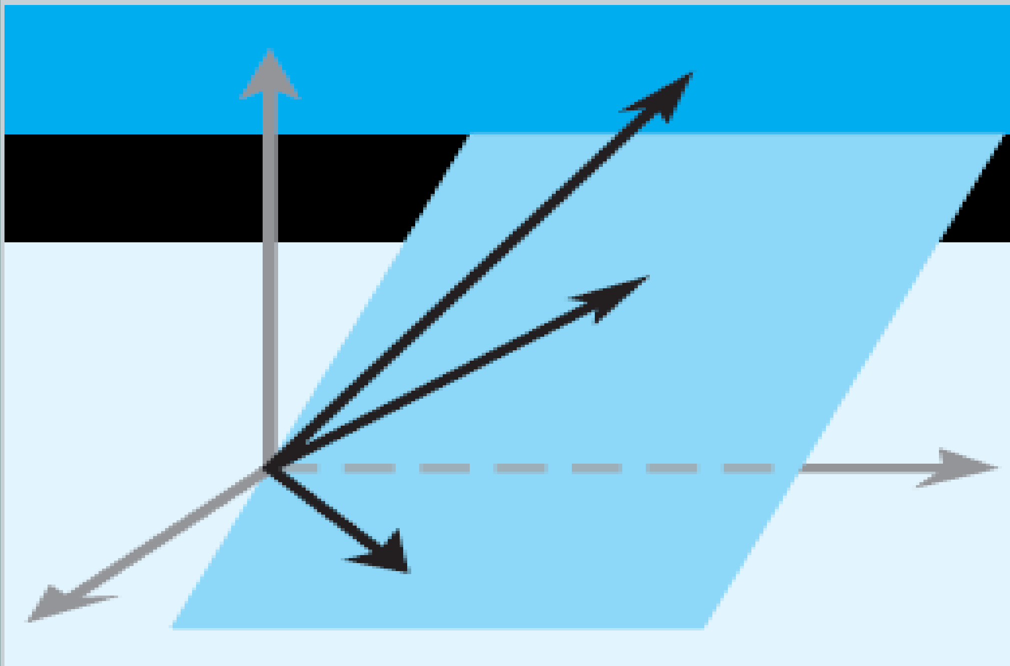
Elementary Linear Algebra



Chapter 5

HOWARD ANTON

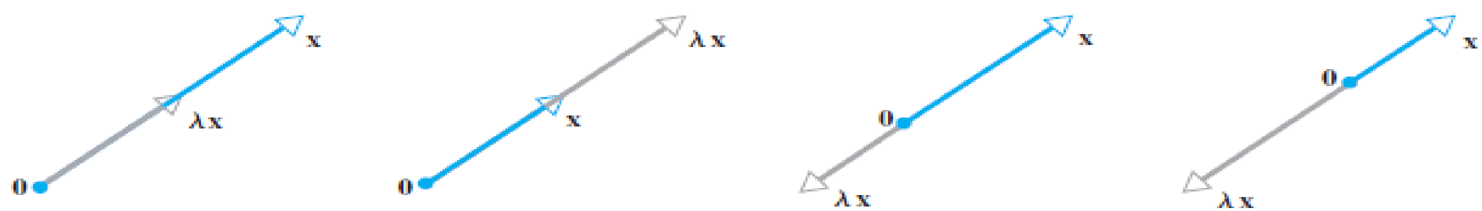
Chapter 5 Eigenvalues and Eigenvectors

Section 5.1 Eigenvalues and Eigenvectors

DEFINITION 1 If A is an $n \times n$ matrix, then a nonzero vector $\mathbf{x}$ in R^n is called an *eigenvector* of A (or of the matrix operator T_A) if $A\mathbf{x}$ is a scalar multiple of $\mathbf{x}$; that is,

$$A\mathbf{x} = \lambda\mathbf{x}$$

for some scalar λ. The scalar λ is called an *eigenvalue* of A (or of T_A), and $\mathbf{x}$ is said to be an *eigenvector corresponding to* λ.



(a) $0 \le \lambda \le 1$ (b) $\lambda \le 1$ (c) $-1 \le \lambda \le 0$ (d) $\lambda \le -1$

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EXAMPLE 1 Eigenvector of a 2 × 2 Matrix

The vector $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is an eigenvector of

$$A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$$

corresponding to the eigenvalue $\lambda = 3$, since

$$A\mathbf{x} = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix} = 3\mathbf{x}$$

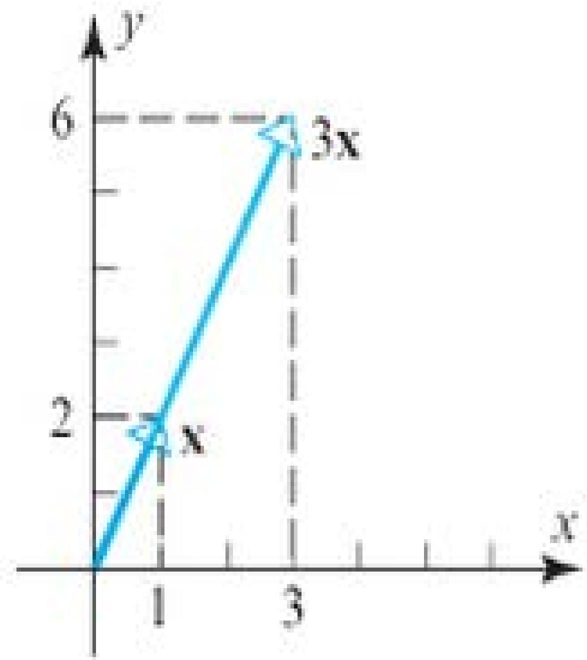

Figure 5.1.2

Geometrically, multiplication by A has stretched the vector $\mathbf{x}$ by a factor of 3 (Figure 5.1.2).

Section 5.1 Eigenvalues and Eigenvectors

THEOREM 5.1.1 *If A is an n x n matrix, then λ is an eigenvalue of A if and only if it satisfies the equation*

$$\det(\lambda I - A) = 0 \tag{1}$$

This is called the characteristic equation of A.

$$\boldsymbol{A}\mathbf{x} = \lambda\mathbf{x}$$

$$\Rightarrow \quad A\mathrm{x} - \lambda\,\mathrm{x} = 0 \quad \Rightarrow \quad A\mathrm{x} - \lambda I\,\mathrm{x} = 0$$

$$\Rightarrow \quad (A - \lambda I)\,\mathrm{x} = 0 \quad \Rightarrow \quad (\lambda I - A)\,\mathrm{x} = 0$$

EXAMPLE 2 Finding Eigenvalues

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In Example 1 we observed that $\lambda = 3$ is an eigenvalue of the matrix

$$A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$$

but we did not explain how we found it. Use the characteristic equation to find all eigenvalues of this matrix.

Solution

It follows from Formula 1 that the eigenvalues of A are the solutions of the equation $\det(\lambda I - A) = 0$, which we can write as

$$\begin{vmatrix} \lambda - 3 & 0 \\ -8 & \lambda + 1 \end{vmatrix} = 0$$

from which we obtain

$$(\lambda - 3)(\lambda + 1) = 0 \qquad (2)$$

Section 5.1 Eigenvalues and Eigenvectors

This shows that the eigenvalues of A are $\lambda = 3$ and $\lambda = -1$. Thus, in addition to the eigenvalue $\lambda = 3$ noted in Example 1, we have discovered a second eigenvalue $\lambda = -1$.

Section 5.1 Eigenvalues and Eigenvectors

EXAMPLE 3 Eigenvalues of a 3 × 3 Matrix **Page 295**

Find the eigenvalues of

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{bmatrix}$$

Solution The characteristic polynomial of A is

$$\det(\lambda I - A) = \det \begin{bmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ -4 & 17 & \lambda - 8 \end{bmatrix} = \lambda^3 - 8\lambda^2 + 17\lambda - 4$$

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EXAMPLE 3 Eigenvalues of a 3 × 3 Matrix

The eigenvalues of ***A*** must therefore satisfy the cubic equation

$$\lambda^3 - 8\lambda^2 + 17\lambda - 4 = 0 \qquad \textbf{(4)}$$

To solve this equation, we will begin by searching for integer solutions. This task can be simplified by exploiting the fact that all integer solutions (if there are any) of a polynomial equation with *integer coefficients*

$$\lambda^n + c_1\lambda^{n-1} + \cdots + c_n = 0$$

Section 5.1 Eigenvalues and Eigenvectors

EXAMPLE 3 Eigenvalues of a 3 × 3 Matrix **Page 295**

must be divisors of the constant term, c_n. Thus, the only possible integer solutions of (4) are the divisors of –4, that is, ±1, ±2, ±4. Successively substituting these values in (4) shows that $\lambda = 4$ is an integer solution. As a consequence, $\lambda - 4$ must be a factor of the left side of (4). Dividing $\lambda - 4$ into $\lambda^3 - 8\lambda^2 + 17\lambda - 4$

$$\begin{array}{r|l} \lambda-4 & \overline{\lambda^3-8\lambda^2+17\lambda-4} \end{array}$$

$$\begin{array}{rl} & \lambda^2 \\ \lambda-4 \big) & \overline{\lambda^3-8\lambda^2+17\lambda-4} \end{array}$$

$$\begin{array}{rl} & \lambda^2 \\ \lambda-4 \big) & \overline{\lambda^3-8\lambda^2+17\lambda-4} \\ & \underline{\lambda^3-4\lambda^2} \\ & \quad -4\lambda^2 \end{array}$$

$$\begin{array}{rl} & \lambda^2-4\lambda \\ \lambda-4 \big) & \overline{\lambda^3-8\lambda^2+17\lambda-4} \\ & \underline{\lambda^3-4\lambda^2} \\ & \quad \underline{-4\lambda^2+16\lambda} \\ & \qquad \lambda-4 \end{array}$$

Section 5.1 Eigenvalues and Eigenvectors

EXAMPLE 3 Eigenvalues of a 3 × 3 Matrix

shows that (4) can be rewritten as $(\lambda-4)\left(\lambda^2-4\lambda+1\right)=0$

Thus, the remaining solutions of (4) satisfy the quadratic equation

$$\lambda^2-4\lambda+1=0$$

which can be solved by the quadratic formula. Thus, the eigenvalues of ***A*** are

$$\lambda=4,\quad \lambda=2+\sqrt{3},\quad \text{and}\quad \lambda=2-\sqrt{3}$$

Section 5.1 Eigenvalues and Eigenvectors

Review:

The general form of a quadratic equation is the following:

$$ax^2 + bx + c = 0$$

The solution(s) to a quadratic equation can <u>always</u> be calculated using the **Quadratic Formula**:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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EXAMPLE 4 Eigenvalues of an Upper Triangular Matrix

Find the eigenvalues of the upper triangular matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{bmatrix}$$

Example 4 eigenvalues of an upper triangular matrix

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Solution

Recalling that the determinant of a triangular matrix is the product of the entries on the main diagonal (Theorem 2.1.2), we obtain

$$\det(\lambda I - A) = \det\begin{bmatrix} \lambda - a_{11} & -a_{12} & -a_{13} & -a_{14} \\ 0 & \lambda - a_{22} & -a_{23} & -a_{24} \\ 0 & 0 & \lambda - a_{33} & -a_{34} \\ 0 & 0 & 0 & \lambda - a_{44} \end{bmatrix}$$

$$= (\lambda - a_{11})(\lambda - a_{22})(\lambda - a_{33})(\lambda - a_{44})$$

Example 4 eigenvalues of an upper triangular matrix

Thus, the characteristic equation is

$$(\lambda - a_{11})(\lambda - a_{22})(\lambda - a_{33})(\lambda - a_{44}) = 0$$

and the eigenvalues are

$$\lambda = a_{11}, \quad \lambda = a_{22}, \quad \lambda = a_{33}, \quad \lambda = a_{44}$$

which are precisely the diagonal entries of A.

Section 5.1 Eigenvalues and Eigenvectors

THEOREM 5.1.2 *If* $\mathbf{A}$ *is an* $n \times n$ *triangular matrix (upper triangular, lower triangular, or diagonal), then the eigenvalues of* $\mathbf{A}$ *are the entries on the main diagonal of* $\mathbf{A}$.

EXAMPLE 5 Eigenvalues of a Lower Triangular Matrix

By inspection, the eigenvalues of the lower triangular matrix

$$A = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ -1 & \frac{2}{3} & 0 \\ 5 & -8 & -\frac{1}{4} \end{bmatrix}$$

are $\lambda = \frac{1}{2}$, $\lambda = \frac{2}{3}$, and $\lambda = -\frac{1}{4}$.

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THEOREM 5.1.3 *If* $\boldsymbol{A}$ *is an* $n \times n$ *matrix, the following statements are equivalent.*

(a) λ *is an eigenvalue of* $\boldsymbol{A}$.

(b) The system of equations $(\lambda \boldsymbol{I} - \boldsymbol{A})\mathbf{x} = \mathbf{0}$ *has nontrivial solutions.*

(c) There is a nonzero vector $\mathbf{x}$ *such that* $\boldsymbol{A}\mathbf{x} = \lambda\mathbf{x}$.

(d) λ *is a solution of the characteristic equation* $\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = 0$.

EXAMPLE 6 Bases for Eigenspaces Page 297

Find bases for the eigenspaces of the matrix

$$A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$$

Solution In Example 1 we found the characteristic equation of $\boldsymbol{A}$ to be

$$(\lambda - 3)(\lambda + 1) = 0$$

from which we obtained the eigenvalues $\lambda = 3$ and $\lambda = -1$.

Thus, there are two eigenspaces of $\boldsymbol{A}$, one corresponding to each of these eigenvalue .

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EXAMPLE 6 **Bases for Eigenspaces**

By definition,

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

is an eigenvector of $\boldsymbol{A}$ corresponding to an eigenvalue λ if and only if **x** is a nontrival solution of $(\lambda \boldsymbol{I} - \boldsymbol{A})\mathbf{x} = 0$, that is, of

$$\begin{bmatrix} \lambda - 3 & 0 \\ -8 & \lambda + 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

EXAMPLE 6 Bases for Eigenspaces

If $\lambda = 3$, then this equation becomes

$$\begin{bmatrix} 0 & 0 \\ -8 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

whose general solution is $x_1 = \frac{1}{2}t, \quad x_2 = t$

(verify) or in matrix form,

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}t \\ t \end{bmatrix} = t \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$$

Thus,

Section 5.1 Eigenvalues and Eigenvectors

EXAMPLE 6 Bases for Eigenspaces

$$\begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$$

is a basis for the eigenspace corresponding to $\lambda = 3$.

We leave it as an exercise for you to follow the pattern of these computations and show that

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

is a basis for the eigenspace corresponding to $\lambda = -1$.

Section 5.1 Eigenvalues and Eigenvectors

EXAMPLE 7 Eigenvectors and Bases for Eigenspaces

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Find bases for the eigenspaces of

$$A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$$

Solution The characteristic equation of A is $\boldsymbol{\lambda^3 - 5\lambda^2 + 8\lambda - 4 = 0}$, or in factored form, $\boldsymbol{(\lambda - 1)\,(\lambda - 2)^2 = 0}$(verify).

Thus, the distinct eigenvalues of A are $\boldsymbol{\lambda = 1}$ and $\boldsymbol{\lambda = 2}$, so there are two eigenspaces of A.

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Section 5.1 Eigenvalues and Eigenvectors

EXAMPLE 7 Eigenvectors and Bases for Eigenspaces Page 298

$$\det(A-\lambda I)=0$$

$$\det(A-\lambda I)=-\lambda(2-\lambda)(3-\lambda)+2(2-\lambda)$$
$$=(2-\lambda)(-\lambda(3-\lambda)+2)$$
$$=(2-\lambda)(\lambda^2-3\lambda+2)=(2-\lambda)(\lambda-1)(\lambda-2)$$

By definition,

$$\begin{bmatrix} \lambda & 0 & 2 \\ -1 & \lambda-2 & -1 \\ -1 & 0 & \lambda-3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \qquad \textbf{(5)}$$

is an eigenvector of A corresponding to λ if and only if $\mathbf{x}$ is a nontrivial solution of $(\lambda I - A)\mathbf{x}=\mathbf{0}$, or in matrix form,

Section 5.1 Eigenvalues and Eigenvectors

EXAMPLE 7 Eigenvectors and Bases for Eigenspaces

In the case where $\lambda = 2$, Formula (5) becomes

$$\begin{bmatrix} 2 & 0 & 2 \\ -1 & 0 & -1 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Solving this system using Gaussian elimination yields (verify)

$$x_1 = -s, \quad x_2 = t, \quad x_3 = s$$

Thus, the eigenvectors of A corresponding to $\lambda = 2$ are the nonzero vectors of the form

$$\mathbf{x} = \begin{bmatrix} -s \\ t \\ s \end{bmatrix} = \begin{bmatrix} -s \\ 0 \\ s \end{bmatrix} + \begin{bmatrix} 0 \\ t \\ 0 \end{bmatrix} = s\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + t\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Section 5.1 Eigenvalues and Eigenvectors

EXAMPLE 7 Eigenvectors and Bases for Eigenspaces

$$\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \text{ and } \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

are linearly independent (why?), these vectors form a basis for the eigenspace corresponding to $\lambda = 2$.

If $\lambda = 1$, then **(5)** becomes

$$\begin{bmatrix} 1 & 0 & 2 \\ -1 & -1 & -1 \\ -1 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 = -2s, \quad x_2 = s, \quad x_3 = s$$

▶ EXAMPLE 7 Eigenvectors and Bases for Eigenspaces

Thus, the eigenvectors corresponding to $\lambda = 1$ are the nonzero vectors of the form

$$\begin{bmatrix} -2s \\ s \\ s \end{bmatrix} = s\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \quad \text{so that} \quad \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

is a basis for the eigenspace corresponding to $\lambda = 1$.

Section 5.1 Eigenvalues and Eigenvectors

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THEOREM 5.1.4 *If* $\mathbf{k}$ *is a positive integer,* λ *is an eigenvalue of matrix* $\mathbf{A}$*, and* $\mathbf{x}$ *is a corresponding eigenvector, then* λ^k *is an eigenvalue of* $\mathbf{A}^k$ *and* $\mathbf{x}$ *is a corresponding eigenvector.*

Proof

EXAMPLE 8 Powers of a Matrix Page 299

In Example 7 we showed that the eigenvalues of

$$A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$$

are $\boldsymbol{\lambda = 2}$ and $\boldsymbol{\lambda = 1}$, so from Theorem **5.1.4** both $\boldsymbol{\lambda = 2^7 = 128}$ and $\boldsymbol{\lambda = 1^7 = 1}$ are eigenvalues of $\boldsymbol{A^7}$. We also showed that

$$\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

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EXAMPLE 8 Powers of a Matrix

are eigenvectors of A corresponding to the eigenvalue $\lambda = 2$, so from Theorem **5.1.4** they are also eigenvectors of A^7 corresponding to $\lambda = 2^7 = 128$. Similarly, the eigenvector

$$\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

of A corresponding to the eigenvalue $\lambda = 1$ is also an eigenvector of A^7 corresponding to $\lambda = 1^7 = 1$.

Section 5.1 Eigenvalues and Eigenvectors

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THEOREM 5.1.5 *A square matrix **A** is invertible if and only if* $\lambda = 0$ *is not an eigenvalue of **A**.*

Proof

EXAMPLE 9 Eigenvalues and Invertibility

The matrix A in Example 7 is invertible since it has eigenvalues $\lambda = 1$ and $\lambda = 2$, neither of which is zero. We leave it for you to check this conclusion by showing that $\det(A) \neq 0$

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THEOREM 5.1.6 Equivalent Statements

If A is an $n \times n$ matrix, then the following statements are equivalent.

(*a*) A *is invertible.*

(*b*) $A\mathbf{x} = \mathbf{0}$ *has only the trivial solution.*

(*c*) *The reduced row echelon form of* A *is* I_n.

(*d*) A *is expressible as a product of elementary matrices.*

(*e*) $A\mathbf{x} = \mathbf{b}$ *is consistent for every* $n \times 1$ *matrix* $\mathbf{b}$.

(*f*) $A\mathbf{x} = \mathbf{b}$ *has exactly one solution for every* $n \times 1$ *matrix* $\mathbf{b}$.

(*g*) $\det(A) \neq 0$.

(*h*) *The column vectors of* A *are linearly independent.*

(*i*) *The row vectors of* A *are linearly independent.*

(*j*) *The column vectors of* A *span* R^n.

(*k*) *The row vectors of* A *span* R^n.

(*l*) *The column vectors of* A *form a basis for* R^n.

(*m*) *The row vectors of* A *form a basis for* R^n.

(*n*) A *has rank* n.

(*o*) A *has nullity* 0.

(*p*) *The orthogonal complement of the null space of* A *is* R^n.

(*q*) *The orthogonal complement of the row space of* A *is* $\{\mathbf{0}\}$.

(*r*) *The range of* T_A *is* R^n.

(*s*) T_A *is one-to-one.*

(*t*) $\lambda = 0$ *is not an eigenvalue of* A.

Homework

Sec 5.1 **Practice for Quiz on 5/12**

# True/False	Page 301
# 6 (a), (e), (f)	Page 326
# 7 corresponding to # 6 (a), (e), (f)	
# 8 corresponding to # 6 (a), (e), (f)	
# 14, 18, 23,	Page 326, 327
# 27, 28	Page 327